# Model accuracy analysis

Portable metric summary. Open model_analysis.html for all SVG charts and full details.

model_name: majority_baseline
split: test
split_role: ID-disjoint partial-start-date subgroup audit
n: 50139
classification.accuracy: 0.312033
classification.balanced_accuracy: 0.25
classification.macro_precision: 0.0780081
classification.macro_recall: 0.25
classification.macro_f1: 0.118912
classification.multiclass_brier: 1.37593
classification.log_loss: 24.7969

model_name: majority_baseline
split: validation
split_role: tuning
n: 14634
classification.accuracy: 0.290898
classification.balanced_accuracy: 0.25
classification.macro_precision: 0.0727245
classification.macro_recall: 0.25
classification.macro_f1: 0.112673
classification.multiclass_brier: 1.4182
classification.log_loss: 25.5586

model_name: phase_logistic
split: test
split_role: ID-disjoint partial-start-date subgroup audit
n: 50139
classification.accuracy: 0.664752
classification.balanced_accuracy: 0.652497
classification.macro_precision: 0.667559
classification.macro_recall: 0.652497
classification.macro_f1: 0.657397
classification.multiclass_brier: 0.449799
classification.log_loss: 0.823993

model_name: phase_logistic

split: validation

split_role: tuning

n: 14634

classification.accuracy: 0.705275

classification.balanced_accuracy: 0.678225

classification.macro_precision: 0.689139

classification.macro_recall: 0.678225

classification.macro_f1: 0.681398

classification.multiclass_brier: 0.406048

classification.log_loss: 0.754162

model_name: phase_refinement_v1

split: test

split_role: ID-disjoint partial-start-date subgroup audit

n: 50139

classification.accuracy: 0.679312

classification.balanced_accuracy: 0.673369

classification.macro_precision: 0.674953

classification.macro_recall: 0.673369

classification.macro_f1: 0.674002

classification.multiclass_brier: 0.433404

classification.log_loss: 0.788326

model_name: phase_refinement_v1

split: validation

split_role: tuning

n: 14634

classification.accuracy: 0.724682

classification.balanced_accuracy: 0.699751

classification.macro_precision: 0.706258

classification.macro_recall: 0.699751

classification.macro_f1: 0.70149

classification.multiclass_brier: 0.384244

classification.log_loss: 0.704155

Retrospective descriptive research decision support; no clinical validation or guaranteed future accuracy.

Tuning data support model selection; calibration data fit transforms/intervals; only test data provide the untouched held-out estimate. Test inspection can influence subsequent experiments.

Metrics are recomputed from every stored prediction. Saved aggregate metrics are parsed and hashed but are not treated as evidence of accuracy.

Current registry snapshots may contain post-start updates and leakage. Completed-only and convenience samples can create selection bias; chronological splits do not establish prospective validity.

No regression accuracy percentage is defined. Duration errors use days. Binary decisions use probability >= 0.5; confusion matrices have actual rows and predicted columns.

Undefined values are null, displayed as unavailable. One-class splits do not identify class discrimination. Small held-out samples and imbalance can make estimates unstable; no uncertainty interval is invented.

Reliability uses ten equal-width bins. Raw and post-hoc calibrated probabilities are analyzed separately; calibration-cohort diagnostics can be optimistic.

Duration-by-phase subgroup metrics cannot be recomputed because saved duration prediction rows do not contain verified phase labels; stored subgroup aggregates are not substituted.

Charts show all stored points (overlapping points can obscure sample counts); metrics and bin counts include every row.

This task predicts recorded phase from protocol characteristics; it does not predict progression or trial success.

Combined, NA and unknown phases are excluded. Class support can be small or absent, and classes absent from training have zero predicted probability.

All held-out records count, including labels absent from training. Confusion matrices use the five canonical labels (rows actual, columns predicted); macro-F1 averages the union of actual/predicted labels; balanced accuracy averages recall for actual supported classes.

Validation scores are optimistic after hyperparameter selection; untouched test scores are the held-out estimate. No performance improvement is guaranteed.

Redaction removes explicit phase mentions but cannot eliminate indirect phase cues or post-start updates in current registry snapshots.

Temporal ordering does not establish that current registry labels or protocol features existed at trial start; baseline provenance must be established separately.

Retrospective research phase-consistency model. Current reported enrollment is not planned enrollment. Test schema slot identifies an ID-disjoint partial-start-date subgroup audit, not chronological/prospective validation. Original exact-date test was exposed and is not this branch's test. Class probabilities are uncalibrated; no clinical/regulatory phase assignment or success claim.

Conditional trial bootstrap excludes sponsor/site clustering, fitting uncertainty and development selection optimism.

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